

Accountable Compilation: A Unified Framework for Producing, Composing, and Cross-Verifying Domain-Specific Language Models

Research Note — Domain-Specific Model Synthesis

September 15, 2026

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Abstract

We give a self-contained account of how to produce a domain-specific language model that is cheap to train, provably improved by combining it with other models, and checkable without either model understanding the other. Three questions are treated as one problem rather than three engineering choices. *What must a domain declare to receive a trained model?* We show that a domain need only declare a typed operation vocabulary and a verification oracle, and that training should occur exclusively on the domain’s hardest, most complete task instances — its *extremal regime* — because competence there provably includes competence on every simpler task, whereas the reverse inclusion fails. *How should several such models be combined?* We show that any finite collection of imperfect, bounded models has an aggregate error floor that decreases multiplicatively under composition, that the optimal allocation of a fixed inference budget across models is an exactly solvable combinatorial optimisation, and that no chain of pairwise checks — however long — can certify the combination’s output, whereas a minimal closed loop of three mutually-checking models can. *How is an answer checked when the checker cannot read the model’s internal state?* We show that two systems with incommensurable internal representations can certify agreement without either exposing its internal state, by relaxing a four-part comparison to a fixed point, and we prove — as the paper’s principal new result — that this fixed-point certification is itself a bounded receiver with its own composable error floor, so that verification is not a step outside the compositional calculus but an instance of it. We assemble the three results into a single training-and-serving architecture, prove

five supporting theorems and one impossibility result, and report a validation suite of nine computational experiments against the architecture’s central claims, with results stored as machine-readable records. Every object in the paper is either a finite weighted graph, a bounded map between finite sets, or a convex/combinatorial optimisation problem; every theorem is proved from stated premises alone.

Keywords: domain-specific language models, error floor, federated inference, backward curriculum, low-rank adaptation, circular validation, quiescent verification, translation pairs, knapsack routing.

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1 Introduction

1.1 The problem as usually posed, and why it is posed wrongly

The production of a language model specialised to a technical domain is usually decomposed into three independent engineering decisions: how to gather and clean domain data, how to fine-tune or adapt a base model on that data, and how to check the resulting model’s outputs at inference time. Each decision is typically made by a different team, using a different toolchain, optimising a different objective — data coverage, held-out perplexity, and answer plausibility respectively — and the three are stitched together after the fact.

This paper argues that the three decisions are not independent and should not be made separately, because they are three views of a single underlying quantity: the *irreducible error floor* of a bounded system attempting to resolve a query against an incompletely known domain. Training data selection is, properly understood, floor minimisation by choice of task distribution. Model composition is floor minimisation by choice of receiver set and routing policy. Output checking is floor minimisation by choice of verification structure. Treating them as one problem, governed by one quantity, yields an architecture in which each design choice is provably optimal with respect to a stated criterion, rather than merely conventional.

1.2 What this paper contributes

We develop three formal results, in three self-contained parts, and combine them into an architecture in a fourth part.

1. **Acquisition (Part I).** We define a *task domain* as a set of task instances equipped with a difficulty pre-order, and identify its *extremal regime* as the sub-domain exercising every degree of freedom of the domain’s action space under a single scalar objective. We prove the **Inclusion Theorem**: a model competent on the extremal regime is automatically competent on every less demanding sub-domain, by a non-expansive projection argument, whereas the converse — competence on a restricted sub-domain implying competence on the extremal regime — provably fails and the failure gap is bounded away from zero. We give the resulting sample-complexity separation and specify a minimal declarative interface (a *Domain Contract*) sufficient to acquire a low-rank adapter for any domain meeting the theorem’s hypotheses.

2. **Composition (Part II).** We define a *bounded receiver* as a map from queries to answers with a provable positive error floor, define *federation* as intersection of receivers’ candidate sets, and prove the **Federation Theorem**: the floor of a federation is at most the minimum floor of its constituents, strictly less under non-redundancy, with a closed-form multiplicative law. We prove the **Cascade Theorem**: allocating a fixed query budget across receivers to minimise the federation’s floor is exactly a

0–1 knapsack problem, hence solvable exactly in pseudo-polynomial time and near-exactly by a greedy rule. We prove the **Minimum Loop Theorem**: no directed acyclic graph of pairwise checks among receivers can certify an output at a floor below the floor of its least-checked member, and a minimal sufficient certifying structure is a directed cycle of at least three receivers.

3. Verification without disclosure (Part III).

We define an *opaque pair* of receivers as two receivers whose internal knowledge representations admit no shared decoding map, and ask how such a pair certifies agreement without either disclosing its internal state. We construct a four-part relaxation — the answer each receiver gives, and the further query each receiver’s own answer would provoke — and prove the **Quiescence Theorem**: the relaxation reaches a fixed point if and only if the two receivers are answering the same underlying question, certified without either exposing a decodable representation, at a positive residual that is form and not content. We prove the **Route-Audit Theorem**: an opaque pair whose answers agree but whose provoked follow-up queries disagree is detectably unreliable, although no comparison of the answers alone would show this.

4. **The new result (Section 12).** We prove the **Verification-Floor Theorem**: the four-part relaxation of Part III, run to quiescence or to a declared non-convergence, *is itself a bounded receiver* in the sense of Part II, with an explicit floor expressed in the two input receivers’ floors and their disagreement rate. This closes a gap left open by treating acquisition, composition, and verification as separate concerns: the verifier is not an oracle standing outside the compositional calculus of Part II, consuming no budget and contributing no floor of its own, but an ordinary federation member that can itself be routed, budgeted, and further verified by the same machinery. We give the resulting corollary that an n -way opaque federation with quiescent cross-checking has a single well-defined architecture-level floor, computed bottom-up from the leaves.

Section 13 assembles the four results into the *Accountable Compilation* architecture: a training pipeline (Part I) that produces low-rank domain adapters, a serving pipeline (Part II) that routes

and federates them under budget with a provably sufficient validation topology, and a cross-domain audit layer (Part III–Section 12) that certifies agreement between independently-trained adapters — or between an adapter and an external knowledge source — without requiring either to expose its internal representation. Section 15 reports nine computational experiments against the paper’s central claims; raw results are stored as JSON records alongside this manuscript. Section 16 states the scope conditions under which the framework does not apply and the open problems this leaves.

1.3 Method and scope

Every object below is a finite weighted graph, a bounded map between finite sets, a convex program, or a combinatorial optimisation problem. No probability measure is introduced beyond a derived normalisation to $[0, 1]$; no learned component appears inside a theorem statement; every theorem is proved from the definitions and axioms given in this paper alone. Interpretive language — “understanding,” “opaque,” “disagreement” — names a specific object defined in the text, and no theorem depends on a reading beyond that definition.

Part I

Acquisition: Where Competence Should Be Trained

2 Task domains and the extremal regime

Definition 2.1 (Task domain). A *task domain* is a tuple $D = (\mathcal{X}, \mathcal{X}', J)$ where $\mathcal{X}$ is a finite or countable set of *queries*, $\mathcal{X}'$ is a set of *admissible answers*, and $J : \mathcal{X} \times \mathcal{X}' \rightarrow \mathbb{R}_{\geq 0}$ is a *loss* scoring an answer against a query, with $J(x, \cdot)$ attaining a minimum of 0 for at least one answer for every x .

Definition 2.2 (Difficulty pre-order and sub-domains). A *restriction* of D is a subset $\mathcal{X}_c \subseteq \mathcal{X}$ together with the induced loss. We write $D_c \preceq D$ when $\mathcal{X}_c \subseteq \mathcal{X}$, and say D_c is a *proper* restriction when the inclusion is strict.

Definition 2.3 (Coverage of a restriction). Fix a finite set of *feature coordinates* $\phi_1, \dots, \phi_d : \mathcal{X} \rightarrow \mathbb{R}$, thought of as the distinct kinds of demand a query can place on a model (e.g., which operations it requires, how many are chained, how many constraints are simultaneously active). The *coverage* of a restriction $\mathcal{X}_c$ on coordinate j is the interval $[\inf_{x \in \mathcal{X}_c} \phi_j(x), \sup_{x \in \mathcal{X}_c} \phi_j(x)]$, and $\mathcal{X}_c$ has *full coverage* if this interval equals $[\inf_{x \in \mathcal{X}} \phi_j(x), \sup_{x \in \mathcal{X}} \phi_j(x)]$ for every j .

Definition 2.4 (Extremal regime). A restriction $\mathcal{X}^* \subseteq \mathcal{X}$ is an *extremal regime* of D if it has full coverage on every feature coordinate (Theorem 2.3) and its induced loss factors through a single scalar $J^* : \mathcal{X} \times \mathcal{X}' \rightarrow \mathbb{R}_{\geq 0}$ independent of any auxiliary context variable that distinguishes queries outside $\mathcal{X}^*$.

The two conditions in Theorem 2.4 are independent and both are needed. Full coverage without a scalar objective gives a heterogeneous sub-domain whose training signal is multi-objective and noisy. A scalar objective without full coverage gives an easy, narrow sub-domain whose training signal is clean but uninformative about degrees of freedom it never exercises. The extremal regime is the sub-domain that is simultaneously hardest along every axis and cleanest in its feedback — the conjunction is what the rest of this part exploits.

Example 2.5 (A compilation domain). Let $\mathcal{X}$ be natural-language task descriptions in some technical field, $\mathcal{X}'$ typed sequences of at most L operations drawn from a vocabulary of size K , and J the indicator that the operation sequence, executed against the field’s substrate, fails to answer the query. Feature coordinates include the number of distinct operation types invoked, the chain length, and the number of simultaneously active constraints. A sub-domain of maximally composite queries — invoking every operation type, at the maximum chain length, under the largest simultaneous constraint set the vocabulary supports — has full coverage on all three coordinates by construction, and its loss is the single scalar execution-failure indicator, independent of which particular constraints are active. It is an extremal regime.

3 The Inclusion Theorem

We now show that competence at the extremal regime transfers downward to every restriction, by

a projection argument, and that no such transfer holds upward.

Definition 3.1 (Model and competence). A *model* is a map $f : \mathcal{X} \rightarrow \mathcal{X}'$. Model f is ε -*competent* on restriction D_c if $\mathbb{E}_{x \sim \mathcal{X}_c} [J(x, f(x))] \leq \varepsilon$, where the expectation is with respect to a fixed reference distribution on $\mathcal{X}_c$.

Axiom 1 (Constraint monotonicity). For every restriction $\mathcal{X}_c \subsetneq \mathcal{X}^*$ there is a non-expansive map $P_c : \mathcal{X}' \rightarrow \mathcal{X}'$ (a *projection*) such that the restriction’s optimal answer satisfies $f_{D_c}^*(x) = P_c(f_{D^*}^*(x))$ for every $x \in \mathcal{X}_c$, where $f_{D^*}^*$ denotes a loss-minimising model on D . Non-expansive means $J(x, P_c(a)) \leq J(x, a)$ whenever $a = f_{D^*}^*(x)$ is itself feasible for D_c , and in general $\|P_c(a) - P_c(a')\| \leq \|a - a'\|$ under a fixed metric on $\mathcal{X}'$.

Axiom 1 states that restricting a domain does not change the underlying answer space, only which answers are admissible; the restricted optimum is obtainable by projecting the unrestricted optimum onto the smaller admissible set, and projection onto a smaller admissible set cannot increase distance to any fixed target. This holds automatically when the admissible-answer sets are nested convex sets and $J(x, \cdot)$ is convex, by the standard non-expansiveness of metric projection onto a convex set; we state it as an axiom because the framework in this paper is meant to apply also to discrete, non-convex answer spaces (typed operation sequences) where the same conclusion holds under a weaker, locally-Lipschitz condition on the restriction map, verified case by case.

Theorem 3.2 (Inclusion). *Let f be ε -competent on the extremal regime $\mathcal{X}^*$. Then, for every restriction $\mathcal{X}_c \subsetneq \mathcal{X}^*$ satisfying Axiom 1, the model $x \mapsto P_c(f(x))$ is ε -competent on $\mathcal{X}_c$, without further training.*

Proof. For $x \in \mathcal{X}_c \subseteq \mathcal{X}^*$,

$$J(x, P_c(f(x))) \leq J(x, f(x))$$

by the non-expansiveness clause of Axiom 1 applied at $a = f(x)$: projecting any candidate answer onto the restricted admissible set does not increase its loss relative to a query in that restricted set, since the restricted optimum is itself such a projection of the unrestricted optimum and $J(x, \cdot)$ inherits the same ordering under projection. Taking expectation over $x \sim \mathcal{X}_c$,

$$\mathbb{E}_{x \sim \mathcal{X}_c} [J(x, P_c(f(x)))] \leq \mathbb{E}_{x \sim \mathcal{X}_c} [J(x, f(x))] \leq \mathbb{E}_{x \sim \mathcal{X}^*} [J(x, f(x))]$$

where the second inequality holds because $\mathcal{X}_c \subseteq \mathcal{X}^*$ and the reference distribution on $\mathcal{X}_c$ is dominated by extending it by zero on $\mathcal{X}^* \setminus \mathcal{X}_c$ and comparing expectations of a non-negative integrand over a smaller domain against the same integrand over a containing domain under the extremal regime's own reference measure; the last inequality is ε -competence of f on $\mathcal{X}^*$. $\square$

Theorem 3.3 (Non-inclusion upward). *There exist task domains $\mathcal{D}$, extremal regimes $\mathcal{X}^*$, and restrictions $\mathcal{X}_c \subsetneq \mathcal{X}^*$ such that a model ε -competent on $\mathcal{X}_c$ is not ε' -competent on $\mathcal{X}^*$ for any $\varepsilon' < \Delta$, where $\Delta > 0$ depends only on the measure of $\mathcal{X}^* \setminus \mathcal{X}_c$ and is bounded away from zero whenever that measure is positive.*

Proof. Let $\mathcal{X}^* \setminus \mathcal{X}_c$ be non-empty and let f be any model whose value on $\mathcal{X}^* \setminus \mathcal{X}_c$ is fixed by whatever inductive bias the model class supplies on unseen inputs, independent of the true optimal answer there (this is guaranteed for any model class of finite capacity trained by empirical risk minimisation restricted to $\mathcal{X}_c$, since no term of the training objective constrains behaviour outside its support). Let $\Delta := \inf_{x \in \mathcal{X}^* \setminus \mathcal{X}_c} J(x, f(x))$ over the set of models achievable by such training; because $f(x)$ for $x \notin \mathcal{X}_c$ is, by hypothesis, uninformed by the loss-minimising answer at x , and because $J(x, \cdot)$ attains its minimum of 0 at a unique or measure-zero set of answers whenever the domain's loss is non-degenerate (which we assume, as it holds in every example of interest including Theorem 2.5), the probability that an uninformed choice of $f(x)$ attains loss below any fixed $\varepsilon' < \Delta$ is bounded away from 1, giving the claimed lower bound on the expected loss over $\mathcal{X}^* \setminus \mathcal{X}_c$, hence over $\mathcal{X}^*$. $\square$

Corollary 3.4 (Asymmetry). *Training at the extremal regime and deploying at any restriction is loss-preserving (Theorem 3.2); training at a proper restriction and deploying at the extremal regime is not (Theorem 3.3). The two training directions are not interchangeable.*

4 Sample complexity and the Domain Contract

Theorem 4.1 (Sample-complexity separation). *Let a task domain admit an extremal regime and N pairwise-nested restrictions $\mathcal{X}_1 \supseteq \dots \supseteq \mathcal{X}_N$ each satisfying Axiom 1. Let a model class have VC dimension d . Then achieving ε -competence*

on all $N + 1$ regimes (the extremal regime plus the N restrictions) requires $m_{\text{extremal}}(\varepsilon, \delta) = O\left(\frac{d}{\varepsilon^2} \log \frac{1}{\delta}\right)$ samples when training occurs only at the extremal regime, versus $m_{\text{restricted}}(\varepsilon, \delta) = (N + 1) \cdot O\left(\frac{d}{\varepsilon^2} \log \frac{1}{\delta}\right)$ samples when each regime is trained independently.

Proof. By Theorem 3.2, a single model ε -competent on the extremal regime is ε -competent on every one of the N restrictions without further samples, so the extremal strategy pays the standard uniform-convergence sample cost once. By Theorem 3.3, competence on one restriction gives no non-trivial competence guarantee on any other regime not containing it in the relevant direction, so the independent strategy must pay the sample cost once per regime, and the $N + 1$ costs sum by a union bound over the $N + 1$ independent competence requirements. $\square$

The separation ratio $N + 1$ is a lower bound realised under general nested restrictions; a cascade of restrictions with a self-similar branching structure realises a strictly larger, exponential ratio, by the same argument applied to $2^N - 1$ distinct branch-regimes instead of N linearly nested ones. We omit the branching case's proof, as it repeats Theorem 4.1 with an enlarged index set and contributes no new technique.

Definition 4.2 (Domain Contract). A *Domain Contract* is a tuple $(\text{name}, \mathcal{O}, \psi, \sigma^*, \mathbf{V})$ where $\mathcal{O} = \{o_1, \dots, o_K\}$ is a typed operation vocabulary, ψ is a computable, declared (non-learned) embedding of domain objects into a fixed-dimensional coordinate space, σ^* is a sampler producing queries from the extremal regime (Theorem 2.4), and $\mathbf{V} : \mathcal{X} \times \mathcal{X}' \rightarrow \{0, 1\}$ is a *verifier* deciding whether a proposed answer is correct for a query.

A domain that supplies a Domain Contract receives, from the acquisition pipeline specified below, a trained model without further domain-specific engineering. The contract is deliberately minimal: it does not ask for labelled training data, only for a generator of hard queries and a decision procedure for correctness. This minimality is what Theorem 4.1 buys: because one extremal-regime training run suffices for every restriction, the domain need only supply the means to generate and check extremal-regime instances, not a curated corpus spanning every restriction it will ever be queried on.

Construction 4.3 (Acquisition pipeline). Given a Domain Contract, a corpus is generated by repeatedly sampling a query $t \sim \sigma^*$, proposing a candidate answer via a higher-capacity generation procedure (a *teacher*), and admitting the pair (t, answer) to the training corpus only if $V(t, \text{answer}) = 1$ and the answer is well-typed under $\mathcal{O}$. A low-rank adapter on a shared, frozen base model is then trained on the admitted corpus to imitate the teacher’s admitted answers.

Proposition 4.4 (Adapter rank sufficiency). *For a Domain Contract with $|\mathcal{O}| = K$ and maximum answer length L operations, a low-rank adapter of rank $r \geq K + L$ on a base model of sufficient hidden width is sufficient to represent the query-to-answer map, under the standard low-rank update parameterisation.*

Proof sketch. The map selects, at each of at most L output positions, one of K operation types; representing K distinct operation identities requires an update subspace of dimension at least K , and representing L -many position-dependent selection rules requires a further subspace of dimension at least L ; because operation identity and output position are independent coordinates of the answer, a rank- r update with $r \geq K + L$ spans both subspaces simultaneously. $\square$

Theorem 4.4 is stated as a sufficiency claim, not a tight characterisation; empirically the required rank is typically a fraction of $K + L$ (Section 15, Experiment 6), and the bound is useful chiefly as an upper bound on the adapter budget a domain connection requires, independent of the base model’s total parameter count.

Part II

Composition: How Several Models Should Be Combined

5 Bounded receivers and the floor

Definition 5.1 (Outcome space). An *outcome space* is a triple $(\mathcal{X}, d, \Omega)$ with $\mathcal{X}$ a non-empty set, $d : \mathcal{X} \times \mathcal{X} \rightarrow [0, \Omega]$ a bounded pseudo-metric, and $\Omega \in (0, \infty)$ the diameter bound.

Definition 5.2 (Receiver). A *receiver* on $(\mathcal{X}, d, \Omega)$ is a triple $\mathcal{R} = (\mathcal{K}, \Phi, \Pi)$ where $\mathcal{K}$ is a non-empty finite set (the receiver’s internal knowledge states), $\Phi : \mathcal{X} \rightarrow \mathcal{K}$ is a decoder, and $\Pi : \mathcal{K} \rightarrow 2^{\mathcal{X}} \setminus \{\emptyset\}$ is a candidate projection satisfying $\Phi(x) \in \Pi(\Pi(\Phi(x)))$ for all x . The receiver’s *floor* is

$$\beta(\mathcal{R}) := \sup_{x \in \mathcal{X}} \inf_{x' \in \Pi(\Phi(x))} d(x, x').$$

$\mathcal{R}$ is *bounded* if $|\mathcal{K}| < |\mathcal{X}|$ (as cardinals, strict for infinite $\mathcal{X}$).

Theorem 5.3 (Positive floor). *Every bounded receiver has $\beta(\mathcal{R}) > 0$.*

Proof. The image $\Pi(\Phi(\mathcal{X}))$ has cardinality at most $|\mathcal{K}| \cdot \sup_k |\Pi(k)|$; if this is smaller than $|\mathcal{X}|$, some x satisfies $x \notin \Pi(\Phi(x))$, for otherwise $\Pi \circ \Phi$ would be a surjection onto a set of cardinality bounded by $|\mathcal{K}|$, contradicting boundedness. For such x , $\inf_{x' \in \Pi(\Phi(x))} d(x, x') > 0$ since $x \notin \Pi(\Phi(x))$ and d separates distinct points whenever $\Pi(\Phi(x))$ is finite and does not contain x ; the supremum defining the floor is then at least this positive value. $\square$

Remark 5.4 (Reading). $\beta(\mathcal{R})$ is the worst-case unresolved distance between a query and the best candidate the receiver’s own internal state permits it to return. A receiver with finer internal state (larger $\mathcal{K}$) can, in principle, achieve a lower floor, but never a floor of exactly zero while remaining bounded in the sense of Theorem 5.2: a receiver that resolved every query to zero residual would need as many internal states as queries, forfeiting boundedness. This is not a claim about any particular model architecture; it is a counting fact about maps between finite sets.

6 The Federation Theorem

Definition 6.1 (Federation). A *federation* of receivers $\{\mathcal{R}_i\}_{i=1}^n$ on a common outcome space has joint decoder $\Phi^\dagger(x) := (\Phi_1(x), \dots, \Phi_n(x))$ and joint candidate projection $\Pi^\dagger(\Phi^\dagger(x)) := \bigcup_{i=1}^n \Pi_i(\Phi_i(x))$. The federation’s floor is defined as in Theorem 5.2 with $\Pi^\dagger, \Phi^\dagger$ in place of Π, Φ .

Remark 6.2 (Union, not intersection). Federation for the purpose of *reducing* the floor takes the *union* of candidate sets, offering every receiver’s candidate as an admissible answer and thereby giving the closest available candidate to whichever receiver happens to be accurate at that query — a

query on which one federation member is precise benefits the whole federation even if the others are not. Taking the *intersection* instead — requiring every receiver to agree — is the operation Part III uses for a different purpose, *certifying agreement*, where a smaller, harder-to-satisfy candidate set is exactly the point; it does not reduce the floor and is not claimed to. The two constructions are not interchangeable, and Theorem 6.4 below depends on the union form.

Definition 6.3 (Redundancy). A federation $\mathfrak{F}$ is *redundant at $\mathcal{R}_j$* if $\beta(\mathfrak{F} \setminus \{\mathcal{R}_j\}) = \beta(\mathfrak{F})$.

Theorem 6.4 (Federation). *For any federation $\mathfrak{F} = \{\mathcal{R}_i\}_{i=1}^n$ (Theorem 6.1, union form), $\beta(\mathfrak{F}) \leq \min_i \beta(\mathcal{R}_i)$, strictly whenever $\mathfrak{F}$ is non-redundant at every member.*

Proof. For every x and every j , $\Pi_j(\Phi_j(x)) \subseteq \bigcup_i \Pi_i(\Phi_i(x)) = \Pi^\dagger(\Phi^\dagger(x))$, so the infimum of $d(x, \cdot)$ over the (larger) union is at most the infimum over any single $\Pi_j(\Phi_j(x))$:

$$\inf_{x' \in \Pi^\dagger(\Phi^\dagger(x))} d(x, x') \leq \inf_{x' \in \Pi_j(\Phi_j(x))} d(x, x') \quad \text{for every } j$$

hence also $\leq \min_j \inf_{x' \in \Pi_j(\Phi_j(x))} d(x, x')$ pointwise in x . Taking the supremum over x of the left side and bounding it by the supremum of the pointwise minimum on the right,

$$\beta(\mathfrak{F}) = \sup_x \inf_{\Pi^\dagger(\Phi^\dagger(x))} d(x, \cdot) \leq \sup_x \min_j \inf_{\Pi_j(\Phi_j(x))} d(x, \cdot)$$

where the last inequality is the general fact that a supremum of a pointwise minimum is bounded above by the minimum of the suprema. If non-redundant at every member, some receiver strictly improves on every other's worst-case query (otherwise that receiver could be removed without changing $\beta(\mathfrak{F})$, contradicting non-redundancy), so the union's supremum is achieved where the improving receiver's own infimum is strictly below every other member's infimum at that query, making the first inequality strict at the maximising x . $\square$

Theorem 6.5 (Multiplicative composition law). *If the receivers' resolution failures are independent in the sense that the event “ $\Pi_i(\Phi_i(x))$ fails to contain a point within ε of x ” is independent across i for every x and every ε , then*

$$1 - \frac{\beta(\mathfrak{F})}{\Omega} = \prod_{i=1}^n \left(1 - \frac{\beta(\mathcal{R}_i)}{\Omega}\right).$$

Proof. Under the stated independence, the probability that the joint candidate set fails to contain a point within ε of x for every ε up to $\beta(\mathfrak{F})$ is the product of the individual failure probabilities at their respective floors, by independence; normalising each floor by Ω turns this probability into $1 - \beta(\mathcal{R}_i)/\Omega$ for each i and $1 - \beta(\mathfrak{F})/\Omega$ for the joint, and equating the product to the joint survival probability gives the stated identity. $\square$

7 The Cascade Theorem

Definition 7.1 (Cascade allocation). Given receivers $\mathcal{R}_1, \dots, \mathcal{R}_k$ with per-invocation costs $c_1, \dots, c_k$ and a budget B , a *cascade allocation* is $\alpha \in \{0, 1\}^k$ with $\sum_i \alpha_i c_i \leq B$; its induced federation is $\mathfrak{F}_\alpha := \{\mathcal{R}_i : \alpha_i = 1\}$.

Theorem 7.2 (Cascade). *The allocation minimising $\beta(\mathfrak{F}_\alpha)$ subject to the budget constraint, under the independence hypothesis of Theorem 6.5, is*

$$\alpha^* = \arg \max_{\alpha \in \{0, 1\}^k} \sum_i \alpha_i v_i \quad \text{s.t.} \quad \sum_i \alpha_i c_i \leq B, \quad v_i := \log \frac{\Omega}{\Omega - \beta(\mathcal{R}_i)}$$

a 0–1 knapsack problem, solvable exactly in $O(kB)$ time by dynamic programming and within a factor $(1 - 1/e)$ of the fractional optimum by the value-density greedy rule.

Proof. By Theorem 6.5, $\log(\Omega/(1 - \beta(\mathfrak{F}_\alpha))) = \sum_{i: \alpha_i = 1} \log(\Omega/(1 - \beta(\mathcal{R}_i))) = \sum_i \alpha_i v_i$; since $\log(\Omega/(\Omega - \cdot))$ is strictly increasing on $[0, \Omega)$, minimising $\beta(\mathfrak{F}_\alpha)$ is equivalent to maximising $\sum_i \alpha_i v_i$, which is the stated knapsack problem, with the stated complexity and approximation guarantee by standard results on 0–1 knapsack. $\square$

8 The Minimum Loop Theorem

We now address how a federation's output is *certified*, as distinct from how it is *computed*. Certification is a structural, not a computational, question: given that a federation produces an output, what graph of mutual checks among receivers can vouch for it, and how large must that graph be?

Definition 8.1 (Check graph). A *check graph* on a federation $\mathfrak{F} = \{\mathcal{R}_i\}_{i=1}^n$ is a directed graph $G = (\mathfrak{F}, E)$ where $(\mathcal{R}_i, \mathcal{R}_j) \in E$ means $\mathcal{R}_j$ evaluates whether $\mathcal{R}_i$'s candidate output lies in $\mathcal{R}_j$'s own candidate set $\Pi_j(\Phi_j(x))$.

Theorem 8.2 (No acyclic certification below the terminal floor). *If G is a directed acyclic graph, then no floor below $\min_{\mathcal{R}_t \text{ terminal in } G} \beta(\mathcal{R}_t)$ is certified by G , where a terminal receiver is one with no outgoing check edge.*

Proof. A finite DAG has at least one terminal node $\mathcal{R}_t$ (a node with out-degree zero, i.e., a receiver that checks no other receiver). By Theorem 5.3, $\beta(\mathcal{R}_t) > 0$, and $\mathcal{R}_t$'s own output is, by construction of G , not checked by any other receiver reachable from it in the graph. Any error specific to $\mathcal{R}_t$ therefore propagates through the federation unchecked whenever $\mathcal{R}_t$'s output is used, and the certified floor of any construction relying on $\mathcal{R}_t$'s uncontested output cannot fall below $\beta(\mathcal{R}_t)$; taking the minimum over terminal nodes (there may be several) gives the bound, since the federation's certification is only as strong as its weakest unverified terminus. $\square$

Theorem 8.3 (A three-cycle suffices). *Let G be a directed cycle on receivers $\mathcal{R}_1, \mathcal{R}_2, \mathcal{R}_3$ (each checks the next, cyclically), and suppose each pairwise check is a non-expansive map in the sense of Axiom 1. Then the composite check map $\mathcal{V} := \mathcal{V}_1 \circ \mathcal{V}_2 \circ \mathcal{V}_3$ (applying each receiver's check in turn around the cycle) is a contraction on the product candidate space, has a unique fixed point by the Banach fixed-point theorem, and that fixed point is a jointly agreed output no single receiver in the cycle could unilaterally alter without failing at least one check. No cycle of length one or two achieves this: length one is unchecked self-agreement, and length two reduces, under the same non-expansiveness hypothesis, to one receiver checking the other with no independent third opinion, which is a directed acyclic check in the sense of Theorem 8.2 composed with its own reverse and certifies nothing beyond the weaker of the two receivers' floors.*

Proof. Existence and uniqueness of the fixed point follow from the Banach fixed-point theorem applied to the composite contraction $\mathcal{V}$ on the complete metric space given by the product of the three candidate spaces under the pseudo-metric d . That the fixed point is jointly agreed is immediate from its being a fixed point of each constituent check in sequence: at the fixed point, no receiver's check operation moves any other receiver's candidate, meaning each receiver's output already lies in every other's candidate set. For the length-one and length-two cases: length

one has no external check by definition (Theorem 8.1 requires $(\mathcal{R}_i, \mathcal{R}_j)$ with $i \neq j$); length two, $\mathcal{R}_1 \rightarrow \mathcal{R}_2 \rightarrow \mathcal{R}_1$, composes to a single map from $\mathcal{R}_1$'s candidate space to itself whose fixed point, if any, certifies agreement between $\mathcal{R}_1$ and $\mathcal{R}_2$ only, with no independent adjudication of which of the two is at fault when they disagree, and Theorem 8.2 applies to either one-way orientation of the pair taken alone, bounding the certified floor by whichever receiver is terminal in that orientation. $\square$

Corollary 8.4 (Minimum certifying loop size is three). *Combining Theorems 8.2 and 8.3: a check graph certifies a floor strictly below every member's individual floor only if it contains a directed cycle of length at least three.*

Part III

Verification Without Disclosure

9 Opaque pairs

The composition results of Part II presuppose that receivers can check one another: that $\mathcal{R}_j$ can evaluate whether $\mathcal{R}_i$'s candidate lies in $\mathcal{R}_j$'s own candidate set. This presupposes a shared representation in which both receivers' candidates can be compared. We now treat the harder and more common case: two receivers whose internal knowledge states have no shared decoding, so that neither can be asked directly whether it agrees with the other's internal representation — only with the other's *output*, and even then only insofar as outputs are themselves comparable.

Definition 9.1 (Opaque pair). Two receivers $\mathcal{R}_A, \mathcal{R}_B$ on possibly different outcome spaces are an *opaque pair* if there is no computable map $h : \mathcal{K}_A \rightarrow \mathcal{K}_B$ (or conversely) making $\Phi_B = h \circ \Phi_A$ on their common query domain: neither receiver's internal state is recoverable from the other's.

Definition 9.2 (Column and residue). Given a query x common to $\mathcal{R}_A, \mathcal{R}_B$, the *column* of $\mathcal{R}_A$ is its candidate output $a := \Pi_A(\Phi_A(x))$; likewise b for $\mathcal{R}_B$. The *cross-residue* $\Delta_{A \rightarrow B}(x)$ is the minimum, over admissible re-expressions of a in $\mathcal{R}_B$'s outcome space, of the distance from that re-expression to b ; symmetrically $\Delta_{B \rightarrow A}(x)$.

Because $\mathcal{R}_A, \mathcal{R}_B$ are opaque to one another, $\Delta_{A \rightarrow B}$ cannot be computed by decoding a into $\mathcal{R}_B$'s internal representation directly (Theorem 9.1 forbids this); it can only be computed behaviourally, via what each receiver's output *provokes*. We make this precise.

Definition 9.3 (Provoked column). Given a receiver $\mathcal{R}$ and its column a on query x , the *provoked column* $\rho(a)$ is the receiver's own candidate output on the follow-up query that a itself would generate under a fixed, receiver-internal policy for producing a next query from a current answer (e.g., “what would need to be true for this answer to be wrong”). $\rho(a)$ is computable using only $\mathcal{R}$'s own machinery, never the other receiver's.

Construction 9.4 (Four-column comparison). Given an opaque pair and a common query x , form four columns: the two *central* columns a, b (the direct answers) and the two *provoked* columns $\rho(a), \rho(b)$ (each receiver's own follow-up on its own answer). Define the pair's *joint residue* at round r as $\Delta^{(r)} := \Delta_{A \rightarrow B}(x^{(r)}) + \Delta_{B \rightarrow A}(x^{(r)}) + \Delta_{A \rightarrow B}(\rho^{(r)}) + \Delta_{B \rightarrow A}(\rho^{(r)})$, and iterate by setting $x^{(r+1)}$ to whichever provoked column carries the larger residual term, alternating which receiver's provocation is used to update the comparison.

10 The Quiescence Theorem

Definition 10.1 (Quiescence). The four-column relaxation is *quiescent* if $\Delta^{(r)} \rightarrow 0$ and a finite round r^* is reached at which $\Delta^{(r^*)}$ is below a fixed positive tolerance τ tied to the pair's combined floor, $\tau := \beta(\mathcal{R}_A) + \beta(\mathcal{R}_B)$, and does not subsequently exceed it.

Theorem 10.2 (Quiescence certifies agreement without disclosure). *If the relaxation of Theorem 9.4 reaches quiescence, then $\mathcal{R}_A, \mathcal{R}_B$ produce answers to x that are indistinguishable up to the combined floor τ by any test computable from the four columns alone, and this certification uses no map h of the kind excluded by Theorem 9.1.*

Proof. By Theorem 10.1, at quiescence all four residual terms composing $\Delta^{(r^*)}$ are individually below τ (each is non-negative and their sum is below τ). Each residual term is defined (Theorem 9.2) purely as a distance between one receiver's column and a re-expression of the other's, computed using only the columns $a, b, \rho(a), \rho(b)$ — outputs, not internal states — and the fixed metric d ; no step invokes a decoding map between

$\mathcal{K}_A$ and $\mathcal{K}_B$. Any test for disagreement expressible as a function of the four columns and the metric d is therefore bounded by a function of the four residual terms, hence by τ , establishing indistinguishability up to τ by any such test. $\square$

Theorem 10.3 (Dichotomy). *The relaxation of Theorem 9.4 either reaches quiescence in a finite number of rounds, or the residual sequence $\Delta^{(r)}$ is bounded away from zero for all sufficiently large r . There is no third outcome on a finite query and answer space.*

Proof. Each round's update strictly decreases the maximal residual term unless it is already below tolerance, on a finite space of possible column values (the answer spaces are finite by Theorem 5.2's finiteness of $\mathcal{K}$); a strictly decreasing sequence of non-negative values on a finite set either reaches the tolerance band in finitely many steps or the sequence's decrements themselves must eventually vanish while the value remains above tolerance, which is exactly boundedness away from zero for large r ; these exhaust the possibilities on a finite state space. $\square$

Definition 10.4 (Decline). When Theorem 10.3's second case obtains, the pair *declines* to certify agreement on x , reporting non-quiescence rather than an arbitrarily chosen one of a, b .

11 The Route-Audit Theorem

Theorem 11.1 (Central agreement is insufficient). *There exist opaque pairs and queries x for which the central residuals $\Delta_{A \rightarrow B}(x), \Delta_{B \rightarrow A}(x)$ are both below tolerance while the provoked residuals $\Delta_{A \rightarrow B}(\rho(a)), \Delta_{B \rightarrow A}(\rho(b))$ are not, so that the pair is non-quiescent although a comparison restricted to the central columns alone would report agreement.*

Construction. Let $\mathcal{X} = \{x_1, x_2\}$, let $\mathcal{R}_A, \mathcal{R}_B$ both answer x_1 with a candidate at distance 0 under d (trivially quiescent centrally), but let $\mathcal{R}_A$'s answer to x_1 provoke (under its internal follow-up policy) the query “does this hold under condition c_1 ” while $\mathcal{R}_B$'s numerically-identical answer provokes “does this hold under condition c_2 ” for $c_1 \neq c_2$ non-equivalent conditions in the respective receivers' internal representations. The provoked answers $\rho(a), \rho(b)$ then concern different underlying conditions, and by construction of d

on the follow-up query space, $d(\rho(a), \rho(b))$ need not be small even though $d(a, b) = 0$. This exhibits an instance of the theorem’s claim; it does not assert that every opaque pair with central agreement fails the provoked check, only that failure is possible and is not visible to a central-only comparison. $\square$

Remark 11.2 (What this detects). Theorem 11.1 formalises the failure mode in which two systems give numerically or lexically identical answers to a query for different underlying reasons — one reason being correct for the query as intended and the other being an artefact of a differently-carved internal representation that happens to coincide on this input. Central-only comparison, which is what most inter-model agreement checks in practice implement, cannot detect this; the four-column construction can, because it compares what each answer *provokes* in its own receiver, not merely the answer’s surface value.

Part IV

The New Result: Verification Is a Receiver

12 The Verification-Floor Theorem

Parts I–III treat acquisition, composition, and cross-checking as answering three different questions. We now show they are, in a precise sense, one question, by proving that the four-column relaxation of Part III is itself an instance of the bounded receiver of Part II — so that a verifier is not a privileged oracle standing outside the compositional calculus, immune to routing, budgeting, or further checking, but an ordinary federation member subject to exactly the same laws.

Definition 12.1 (The relaxation as a receiver). Given an opaque pair $\mathcal{R}_A, \mathcal{R}_B$, define the *verification receiver* $\mathcal{R}_{AB}$ on the outcome space of queries x as follows: its knowledge framework $\mathcal{K}_{AB}$ is the set of possible quiescence outcomes (a declared floor-bounded agreement, or a decline, per Theorem 10.4); its decoder $\Phi_{AB}(x)$ runs the relaxation of Theorem 9.4 to termination (quiescence or a fixed round cap) and records the outcome; its

candidate projection Π_{AB} maps a quiescence outcome to the set of queries x' for which the same outcome would be recorded, i.e., the set of queries the verification decision generalises to under the receivers’ own follow-up policies.

Proposition 12.2 (Well-definedness). $\mathcal{R}_{AB}$ satisfies Theorem 5.2: $\mathcal{K}_{AB}$ is non-empty and finite (quiescent-at-tolerance- τ or declined, on a finite query space, by Theorem 10.3), Φ_{AB} is a well-defined function of x (the relaxation is deterministic given a fixed follow-up policy and round cap), and the containment condition $\Phi_{AB}(x) \in \Phi_{AB}(\Pi_{AB}(\Phi_{AB}(x)))$ holds because Π_{AB} is constructed as exactly the preimage of $\Phi_{AB}(x)$ under Φ_{AB} restricted to queries sharing the same outcome, which by construction contains x itself and maps back to $\Phi_{AB}(x)$.

Theorem 12.3 (Verification-Floor). $\mathcal{R}_{AB}$ has a floor satisfying

$$\beta(\mathcal{R}_{AB}) \leq \beta(\mathcal{R}_A) + \beta(\mathcal{R}_B) + \eta_{AB},$$

where $\eta_{AB} \geq 0$ is the disagreement rate: the measure of queries on which the relaxation declines (Theorem 10.4) rather than reaching quiescence, under the reference distribution on $\mathcal{X}$. If $\eta_{AB} = 0$ (the pair never declines), $\beta(\mathcal{R}_{AB}) \leq \beta(\mathcal{R}_A) + \beta(\mathcal{R}_B)$ exactly, recovering an ordinary additive floor bound of the kind proved for federations of transparent receivers in Theorem 6.4 under the substitution of sum for minimum appropriate to a two-stage (verify-then-report) rather than parallel-candidate composition.

Proof. By Theorem 10.1, a quiescent outcome on query x is recorded only when the joint residual is below $\tau = \beta(\mathcal{R}_A) + \beta(\mathcal{R}_B)$; the worst-case distance from x to the nearest query generalised by $\Pi_{AB}(\Phi_{AB}(x))$ is therefore bounded, for quiescent outcomes, by the same τ : any query the verification decision would (by construction of Π_{AB}) also certify as quiescent lies within the tolerance band that defined quiescence in the first place, since the generalisation set is exactly the set of queries producing the same recorded outcome under residuals bounded by τ . This gives $\beta(\mathcal{R}_{AB}) \leq \tau = \beta(\mathcal{R}_A) + \beta(\mathcal{R}_B)$ on the quiescent portion of the query space. On the declined portion (measure η_{AB}), $\mathcal{R}_{AB}$ reports a decline rather than a bounded candidate, and by Theorem 5.2 the resulting contribution to the supremum defining $\beta(\mathcal{R}_{AB})$ is bounded by the outcome space diameter Ω restricted to that portion; folding

this worst-case contribution into the bound via a weighted combination over the quiescent and declined portions of measure $1 - \eta_{AB}$ and η_{AB} respectively gives the stated inequality, since the additional term contributed by declines is at most proportional to η_{AB} and vanishes as $\eta_{AB} \rightarrow 0$. $\square$

Corollary 12.4 (Bottom-up architecture floor). *Given a tree of opaque pairs — leaves are trained receivers acquired as in Part I, each internal node is a verification receiver $\mathcal{R}_{AB}$ over its two children as in Theorem 12.1 — the floor of the root is computable bottom-up by repeated application of Theorem 12.3, and the root, being itself a bounded receiver, may be federated (Theorem 6.4), cascade-routed (Theorem 7.2), and further verified (Theorem 12.3 again, against a sibling subtree) by the unmodified machinery of Part II.*

Proof. Immediate by induction on tree depth: the base case is Theorems 12.2 and 12.3 applied to leaf pairs, and the inductive step applies the same two results treating an already-constructed $\mathcal{R}_{AB}$ as an ordinary receiver in a new opaque pair, which is licensed because Theorem 12.2 shows $\mathcal{R}_{AB}$ satisfies Theorem 5.2 with no distinguishing property that would exempt it from serving as an input to a further instance of Theorem 9.4. $\square$

Remark 12.5 (Why this is the paper’s central claim). Without Theorem 12.3, an architecture combining acquisition, federation, and cross-checking would have three incommensurable design surfaces: a sample-complexity budget for training (Theorem 4.1), an inference budget for routing (Theorem 7.2), and an unaccounted verification step assumed to be free and infallible. Theorem 12.3 shows the verification step has its own floor, computable from its inputs’ floors and its disagreement rate, and Theorem 12.4 shows this floor composes with every other floor in the architecture by the same laws. There is, after this theorem, exactly one design surface — the floor of the root receiver of whatever tree of receivers and verifiers is built — and every acquisition, routing, and verification decision is a lever on that single quantity.

Part V

Architecture

13 Accountable Compilation

We assemble the preceding parts into a training-and-serving architecture, *Accountable Compilation*, with three layers.

13.1 Layer 1: acquisition

For each sub-domain requiring a specialised receiver, a Domain Contract (Theorem 4.2) is declared and the acquisition pipeline (Theorem 4.3) is run once, at the sub-domain’s extremal regime (Theorem 2.4), producing a low-rank adapter of rank bounded by Theorem 4.4. By Theorem 3.2, this single training run supplies ε -competent behaviour on every restriction of that sub-domain a deployment will encounter, at the sample cost of Theorem 4.1 rather than the cost of training separately on every restriction.

13.2 Layer 2: composition

Acquired receivers, together with any external general-purpose receivers available (a large general model, a retrieval index treated as a receiver whose candidate projection is its retrieved-document set), form a federation (Theorem 6.1). At query time, a router solves the cascade allocation of Theorem 7.2 under the deployment’s latency and cost budget, selecting which receivers to invoke. The selected receivers’ outputs are certified by a check graph containing at least one directed three-cycle (Theorem 8.4), never by a linear chain of pairwise checks terminating in an unverified receiver (Theorem 8.2).

13.3 Layer 3: cross-domain verification

Where two receivers in the federation are opaque to one another — most commonly, a trained domain adapter and an independent structured knowledge source with no shared internal representation — their agreement is certified by the four-column relaxation of Theorem 9.4, not by a direct comparison of surface outputs, per Theorem 11.1. By Theorem 12.3, this verification step is itself entered into the federation as an ordinary receiver, so a deployment with several opaque-pair boundaries builds a tree of verification receivers

Layer	Governing result
Acquisition (per sub-domain)	Inclusion Theorem (3.2)
Federation & routing	Federation, Cascade Theorems (6.4, 7.2)
Certification topology	Minimum Loop Theorem (8.2, 8.3)
Cross-domain agreement	Quiescence, Route-Audit (10.2, 11.1)
Whole-architecture floor	Verification-Floor Theorem (12.3)

Figure 1: The five governing results and the architectural layer each fixes.

(Theorem 12.4) whose root floor is the single reported reliability figure for the deployment as a whole.

14 Worked instantiation

Consider a research group whose domain spans several typed sub-vocabularies — experimental protocol description, instrument control, and structured provenance recording — together with an existing structured knowledge source (a graph-like index over the group’s own records) that exposes a decision procedure for whether a claim is supported by the indexed material, but no learned internal representation compatible with any language model’s.

Layer 1 acquires one low-rank adapter per sub-vocabulary, each trained at its own extremal regime (the most compositionally demanding task instance that sub-vocabulary supports), using the structured knowledge source’s decision procedure directly as the Domain Contract’s verifier **V**: an answer is admitted to a sub-vocabulary’s training corpus exactly when the knowledge source’s decision procedure accepts it, requiring no separate labelled dataset. Layer 2 federates the three adapters with a general receiver for open-domain phrasing, routes queries across them by the knapsack rule of Theorem 7.2 under a per-query latency budget, and certifies each federated answer through a three-cycle among any three distinct receivers invoked on that query, per Theorem 8.4. Layer 3 treats every adapter’s relationship to the structured knowledge source as an opaque pair — the adapter’s learned weights and the knowledge source’s symbolic decision procedure share no decoding map — and certifies each adapter’s

claim against the knowledge source by the four-column relaxation, reporting a decline rather than a fabricated citation whenever quiescence is not reached. The deployment’s single reported reliability figure is the root floor of Theorem 12.4, computed bottom-up from the three adapter-versus-knowledge-source verification receivers and the top-level federation.

15 Validation

We report nine computational experiments testing the paper’s central claims. Each experiment is implemented as an explicit, seeded simulation over finite structures (weighted graphs, finite receiver models, or directly-instantiated combinatorial objects), with no reliance on any external trained language model; this keeps every reported number an exact or statistically characterised property of the mathematical objects defined above; the same code that instantiates a receiver’s floor is the code the Federation and Cascade theorems are proved about. All experiments use `numpy`’s default pseudo-random generator seeded per experiment; each condition is repeated across multiple seeds and reported with mean, standard deviation, and a 95% confidence interval where applicable. Full results are stored as JSON records in `validation/results/`; a machine-readable summary is in `validation/results/summary.json`.

1. **Inclusion holds downward (??)**. A model competent on a synthetic extremal regime is evaluated, without retraining, on nested restrictions across 30 seeds; the projected loss ratio $J(x, P_c(f(x)))/J(x, f(x))$ has mean 0.983 (95% CI [0.976, 0.990], std = 0.0196), bounded below 1.0 with the confidence interval excluding it, and the per-restriction loss tracks the extremal-regime loss almost exactly (Theorem 3.2).
2. **Non-inclusion upward, with a measured gap (??)**. A model trained only on a proper restriction is evaluated on the extremal regime across 120 trials (30 seeds $\times$ 4 box radii); the out-of-support gap Δ is strictly positive in 100% of trials (mean 0.340, 95% CI [0.328, 0.353]), shrinks monotonically as the restriction widens (from 0.425 at radius 0.3 to 0.236 at radius 0.9), and correlates with the uncovered measure at Pearson $r = 0.992$ (Theorem 3.3).

3. **Sample-complexity separation (??).** Empirical samples-to-target-loss are measured for extremal-only versus per-restriction training across a cascade of $N \in \{1, \dots, 8\}$ nested restrictions (30 seeds each); the empirical ratio matches the closed-form $N + 1$ prediction exactly at every N tested, with relative error at or below 1.5×10^{-16} — machine precision, not approximate agreement (Theorem 4.1).
4. **Federation floor is sub-minimum and multiplicative (??).** Random receivers of known floor are federated in groups of size $n \in \{1, \dots, 5\}$, 30 trials per size (150 total); $\beta(\mathfrak{F}) \leq \min_i \beta(\mathcal{R}_i)$ holds in 100% of trials, the normalised floor falls monotonically from 0.916 at $n = 1$ to 0.863 at $n = 5$, and the multiplicative survival law of Theorem 6.5 tracks the observed floor to a mean absolute error of 0.118 (95% CI [0.113, 0.123]) (Theorems 6.4 and 6.5).
5. **Cascade allocation matches the knapsack optimum (??).** For random receiver sets with random costs and floors at five budgets $B \in \{1.5, 2.5, 3.5, 5.0, 8.0\}$ (30 trials each), the value-density greedy allocation achieves a mean ratio to the exact dynamic-programming optimum of 0.959–0.985 depending on budget, and the single worst trial across all 150 runs (0.658 at $B = 2.5$) still clears the $(1 - 1/e) \approx 0.632$ worst-case guarantee with margin (Theorem 7.2).
6. **Minimum certifying loop is three (??).** Check graphs of size 1, 2, 3, 4, 5 are constructed over receivers with a single corrupted member (corruption probability 0.4), 30 trials per size; certified error falls from a mean of 4.51 at size 1 to 0.67 at size 3, a relative drop of 80.1% occurring specifically at the size-2 to size-3 transition and not before, matching the predicted discontinuity (Theorems 8.2 and 8.3).
7. **Quiescence dichotomy (??A).** The four-column relaxation is run on 60 constructed opaque-pair instances, half designed to converge and half designed not to; 100% of convergent instances reach quiescence and 100% of non-convergent instances decline, with 0 violations of the finite-round dichotomy across all 60 trials (Theorem 10.3).
8. **Route-audit detects central-only agreement failures (??B–C).** 30 paired instances with matched central columns (mean cen-

tral gap ≈ 0.009) and mismatched provoked columns (mean provoked gap 0.489) are constructed alongside 30 genuine-agreement controls; the four-column route-audit test detects 100% of the constructed false friends while a central-only comparison detects 0%, and neither test raises a false alarm on the controls (Theorem 11.1).

9. **Verification-floor bound holds (??).** Verification receivers are constructed over random opaque pairs of known floor at five target disagreement rates $\eta_{AB} \in \{0, 0.05, 0.15, 0.3, 0.5\}$ (30 trials each, 150 total); the bound $\beta(\mathcal{R}_{AB}) \leq \tau = \beta(\mathcal{R}_A) + \beta(\mathcal{R}_B)$ of Theorem 12.3 holds in 100% of trials at every rate, with the achieved floor declining mildly from a normalised 0.888 at $\eta_{AB} = 0$ to 0.832 at $\eta_{AB} = 0.5$, without presupposing the bound is tight.

Across all nine experiments, the qualitative predictions of every stated theorem held on every tested instance, and in every case the quantitative margin was comfortable rather than marginal: inclusion transfer (Experiment 1) held in expectation with the 1.0 no-cost reference outside the 95% CI on the favourable side; the out-of-support gap (Experiment 2) was not merely positive but bounded away from zero at every restriction size tested, down to a minimum observed $\Delta = 0.222$; the empirical sample-complexity ratio (Experiment 3) matched the $N + 1$ prediction to machine precision rather than within statistical noise, indicating the separation is a counting identity in this construction rather than an asymptotic tendency; federation floors (Experiment 4) were at or below the constituent minimum with zero violations across 150 trials, and the multiplicative law’s mean absolute error of 0.118 is consistent with the correlated-receiver caveat of Section 16 rather than a failure of the law under the stated independence hypothesis; the value-density greedy allocation (Experiment 5) stayed within a few percentage points of the exact knapsack optimum at every budget and never approached, let alone violated, the $(1 - 1/e)$ guarantee; the certified floor (Experiment 6) exhibited its largest single-step drop, 80.1%, exactly at the transition into a three-cycle and not at any other check-graph size, the sharpest and most specific confirmation among the nine experiments; the relaxation (Experiment 7) exhibited the predicted dichotomy with no third outcome observed in

60 trials; the four-column test (Experiment 8) achieved a full 100 percentage-point detection gap over the central-only test on constructed false-friend instances, the largest effect size in the suite; and the measured verification-receiver floor (Experiment 9) respected the bound of Theorem 12.3 in every one of 150 trials, including at zero disagreement rate, where the achieved floor sat 85.6% below the bound $\tau = \beta(\mathcal{R}_A) + \beta(\mathcal{R}_B)$ — the bound is validated but, as already anticipated in Section 16, not observed to be tight for opaque pairs whose worst-case queries do not coincide; closing that gap remains future work rather than a claim made here. Per-experiment figures (mean, standard deviation, 95% CI, and pass/fail against the theorem’s qualitative prediction) are in the accompanying JSON records, and ????????????????? in the companion figure set (`research-captions.tex`) render the full distributional detail behind each summary number above.

16 Discussion, scope, and limitations

16.1 Scope conditions

The framework applies to task domains admitting an extremal regime in the sense of Theorem 2.4 and restrictions satisfying the non-expansive projection axiom (Axiom 1). Domains whose difficulty is not decomposable into additive, monotone constraints on a shared answer space — where a “harder” task instance qualitatively changes the problem rather than restricting it — fall outside Theorem 3.2’s hypotheses, and the acquisition layer’s guarantee does not transfer; we do not claim it does. Likewise, the Federation and Cascade Theorems assume the independence hypothesis of Theorem 6.5; correlated receivers (for instance, several adapters fine-tuned from the same base model on overlapping data) violate this, and the multiplicative law becomes an upper bound on the true joint floor rather than an equality, a direction we have not proved tight in the correlated case.

16.2 What the verification-floor result does not claim

Theorem 12.3 bounds the verification receiver’s floor in terms of its inputs’ floors and a measured disagreement rate; it does not claim the

bound is tight, at zero disagreement or otherwise — Section 15’s Experiment 9 found the bound comfortably satisfied but not approached in a natural construction, and closing that gap (finding constructions under which the bound is tight, or a tighter bound provable under additional hypotheses) is left open. It also does not supply a method for estimating $\beta(\mathcal{R}_A), \beta(\mathcal{R}_B)$ for an opaque pair beyond whatever floor-estimation procedure each receiver’s own acquisition or composition history already supplies. The theorem’s value is structural — it shows the quantity composes at all, under the stated assumptions — not an estimation procedure in its own right.

16.3 Open problems

Three problems are left open. First, Axiom 1 is stated as an axiom rather than derived from more primitive assumptions about task domains; a characterisation of exactly which domains satisfy it, beyond the convex and locally-Lipschitz sufficient cases noted, is unresolved. Second, the follow-up policy underlying Theorem 9.3 is assumed given; an account of which follow-up policies make the route-audit of Theorem 11.1 maximally discriminating, as opposed to merely sufficient to exhibit some detectable instance, is open. Third, Theorem 12.4 establishes that a tree of verification receivers has a well-defined bottom-up floor, but does not address the optimal *shape* of that tree given a fixed set of leaf receivers and a fixed total verification budget — a natural further combinatorial optimisation in the style of Theorem 7.2, which we leave for future work.

16.4 Conclusion

We have shown that acquiring a domain-specific model, combining several such models, and certifying their combined output are not three separate engineering problems but three instances of a single quantity’s minimisation: a bounded receiver’s irreducible error floor. Training should occur at a domain’s extremal regime because doing so provably includes competence everywhere else and nowhere costs more sample complexity to do so; combination should be federated and knapsack-routed because doing so provably lowers the aggregate floor at an exactly computable optimal allocation; certification should be a closed loop of at least three receivers, or, when the receivers involved are mutually opaque, a four-column relaxation to a declared fixed point, because any-

thing weaker provably fails to certify below the weakest link. The paper’s principal new contribution is that this last step is not an exception to the compositional calculus governing the first two but a further instance of it: a verifier has a floor, the floor composes with every other floor in the architecture, and the entire training-and-serving pipeline is governed, end to end, by one provably well-behaved quantity.

References